refinements-changes.md - FLIPSIDE

Running Development Log Developer: Gift Zowa **Student Number:** ST10443108

This log runs across all three days of development. Each entry records scope shifts, design decisions, bugs found and fixed, AI prompts used, art generation notes, and manual changes. Maintained with assistance from Cursor and Claude.

Day 1 - Ideation & Planning

[DAY 1 - ENTRY 1]

Time: Morning **Type:** Design Decision **Entry:** Received theme: "Switch." Brainstormed five concepts: SWITCHBOARD (cable routing puzzle), FLIPSIDE (split-screen dual character), PERSONA (stealth identity swap), SURGE (electricity action), HALFLIFE (ghost/physical platformer). Selected FLIPSIDE as the strongest fit. Initial concept had both characters controlled simultaneously with mirrored input. **AI Contribution:** Claude used to generate and evaluate all five concepts. FLIPSIDE selected by developer.

[DAY 1 - ENTRY 2]

Time: Mid-Morning **Type:** Design Decision - Mechanic Revision **Entry:** Revised the core mechanic. Simultaneous mirroring replaced with relay/switch system: one character active at a time, Tab to switch control, inactive character freezes, physical Switch objects change the other world. The relay system expresses the theme more directly, makes for richer puzzle design, and avoids the frustration of trying to manage both characters at once. **AI Contribution:** Claude used to evaluate revised mechanic directions. Relay system finalised by developer.

[DAY 1 - ENTRY 3]

Time: Late Morning **Type:** Documentation **Entry:** Created all Day 1 documents. Claude assisted with structure and writing. All content reviewed and personalised by developer.

[DAY 1 - ENTRY 4]

Time: Afternoon **Type:** Setup **Entry:** Unity 2022.3 LTS project created (URP 2D). Packages installed: 2D Tilemap Editor, New Input System, Cinemachine, TextMeshPro, 2D Animation. New Input System accepted. Git initialised. Tags created: ARC, ECHO. Layer created: Ground.

[DAY 1 - ENTRY 5]

Type: Setup - Cursor Context **Entry:** Created .cursorrules in project root. Contents: C# Unity 2022.3 LTS, New Input System only, rb.linearVelocity not rb.velocity, active/inactive state pattern, singleton naming convention, SerializeField only (no FindObjectOfType), CompareTag for character identification.

[DAY 1 - ENTRY 6]

Type: Code - AI Assisted **Entry:** PlayerController.cs generated with Cursor. **Prompt:** "Create a Unity 2D PlayerController using Rigidbody2D and the New Input System. Support movement and jump with coyote time (0.15s) and jump buffer (0.1s). Ground detection via BoxCast. isActive bool - false = Kinematic + zero velocity. All values SerializeField. Use rb.linearVelocity." **Assessment:** Functional. Ground layer needed manual setup. Fall gravity multiplier added manually. jumpForce tuned 8 to 11, gravityScale tuned to 2.8. CapsuleCollider2D substituted for BoxCollider2D for better character feel.

[DAY 1 - ENTRY 7]

Type: Code - AI Assisted **Entry:** EchoController.cs generated. **Prompt:** "Inherit from PlayerController with no overrides - ECHO controls identically to ARC, no mirroring." **Assessment:** Clean single-file inheritance. No changes needed.

[DAY 1 - ENTRY 8]

Type: Code - AI Assisted **Entry:** CharacterSwitcher.cs generated. **Prompt:** "Singleton. Tab via Broadcast Messages toggles SetActive on ARC and ECHO. Smooth camera brightness transition - active to 1.0, inactive to 0.45." **Assessment:** Core logic correct. backgroundColor simplification acceptable for jam scope. SwitchToARC/SwitchToECHO methods added manually. GetARC/GetECHO added later for LevelManager freeze system.

[DAY 1 - ENTRY 9]

Type: Code - AI Assisted **Entry:** WorldSwitch.cs and SwitchableObject.cs generated. **WorldSwitch prompt:** "ActivatedBy enum (ARC/ECHO/Either). OnTriggerEnter2D checks tag. Calls OnSwitchActivated() on List of SwitchableObject targets. autoSwitchCharacter bool - triggers ToggleActiveCharacter() after 0.4s." **SwitchableObject prompt:** "Toggle/TurnOn/TurnOff enum. startsActive bool. Collider2D enabled/disabled. Sprite colour

white=active, semi-transparent=inactive." **Assessment:** Both clean. Ghosting approach from AI kept. ResetSwitch/ResetObject added manually.

[DAY 1 - ENTRY 10]

Type: Code - AI Assisted **Entry:** SplitScreenSetup, CameraFollow, HazardTile, ExitDoor, LevelManager, UIManager, LevelCompleteUI, MainMenu generated in one session. Key manual additions: LevelManager auto-switch on exit reached with hint label; UIManager FlashDividingLine() method.

[DAY 1 - ENTRY 11]

Type: Art - AI Assisted **Entry:** First tile sets generated in Leonardo AI. Settings: Leonardo Diffusion XL, Guidance Scale 7, Alchemy ON, 512x512. **ARC ground prompt:** "2D game platform tile, white and pale grey, geometric minimalist, clean sharp edges, grid lines engraved, cool blue tint, pixel art, flat colours, no shading, no gradients, transparent background, game asset" **ECHO ground prompt:** "2D game platform tile, deep black and dark charcoal, geometric minimalist, neon purple glowing edge, pixel art, flat colours, no shading, transparent background, game asset" **Assessment:** First batch was too painterly. Added "extremely simple, 8-bit style, minimal detail" to the prompt. Second batch was clean. Used Image-to-Image to generate ECHO variants from ARC tiles - kept proportions consistent across both worlds. PPU issue: Leonardo output at irregular sizes (297x249, 328x328 etc.). Fixed with birme.net batch resize to 512x512. PPU set to 512 on all tile sprites.

Day 2 - Prototype Build

[DAY 2 - ENTRY 1]

Type: Setup **Entry:** Split-screen cameras configured. ARC world at Y=0, ECHO world at Y=-200. CameraARC follows ARC, CameraECHO follows ECHO. Cameras naturally never see each other's worlds - no layer culling needed. Both cameras Z=-10. Orthographic size 4 for good split-screen view.

[DAY 2 - ENTRY 2]

Type: Bug Fix **Entry:** Tiles appeared too large in Tilemap. Root cause: Leonardo images were non-square irregular sizes. Fixed with birme.net batch resize to 512x512, PPU set to 512.

[DAY 2 - ENTRY 3]

Type: Code - AI Assisted **Entry:** LevelTimer.cs generated. **Prompt:** "Dual countdown timers for ARC and ECHO. Active character drains at 1x speed, inactive at 0.25x. Timer text in each world half. White to orange at 10s, red at 5s. Either timer hitting zero resets the level." **Assessment:** Core logic correct. Frost icon system, vertex wave animation for inactive timer, and strobe effect for critical threshold added manually for visual polish.

[DAY 2 - ENTRY 4]

Type: Level Design **Entry:** Levels 1-4 built in Tilemap editor. Level 1: flat tutorial. Level 2: first WorldSwitch - ARC creates platform for ECHO. Level 3: gaps + switch dependency. Level 4: ECHO deactivates hazard for ARC - first reverse dependency. All tested and functional.

[DAY 2 - ENTRY 5]

Type: Setup **Entry:** Canvas and UIManager set up. All UI anchored correctly using Scale With Screen Size (1920x1080). Timer text: ARC anchored top-right, ECHO anchored bottom-right. Level label anchored top-centre. FlashOverlay stretched full canvas. Dividing line anchored middle-centre with full-width stretch.

[DAY 2 - ENTRY 6]

Type: Art - AI Assisted **Entry:** Character sprites generated in Leonardo AI. **ARC idle prompt:** "2D game character sprite, small humanoid figure, pure white silhouette, simple geometric body, no facial features, standing idle, pixel art, flat colour, no shading, transparent background, 512x512, platformer character" **ECHO idle prompt:** Same as ARC with "deep black silhouette, neon purple glow outline" **Assessment:** Silhouette approach worked well. Used Image-to-Image to generate run/jump variants from idle base - kept proportions consistent. Sliced in Sprite Editor using Grid By Cell Size.

Day 3 - Refinement & Submission

[DAY 3 - ENTRY 1]

Type: Code - AI Assisted - BoostOrb system **Entry:** BoostOrb.cs generated. **Prompt:** "Pickup orb that boosts the OTHER character (not the collector). Speed/Jump/Both types. Duration configurable. Bob and rotate animation. Pop-burst on collect." **Assessment:** Output was good to start with. Found a critical bug during testing: the coroutine ran on the BoostOrb

MonoBehaviour itself. Destroy(gameObject) on collection killed the coroutine before the duration finished, so stats were never restored and the boost lasted forever. Fixed by moving ApplyBoost() coroutine to PlayerController. The orb now calls recipient.ApplyBoost() and destroys itself - the timer runs on the character and always completes.

[DAY 3 - ENTRY 2]

Type: Code - AI Assisted - MovingHazard **Entry:** MovingHazard.cs generated. **Prompt:** "Moving hazard with Horizontal/Vertical/Custom movement types. SineWave and EaseInOut easing. StartDelay. Ghost trail effect - faded sprite copies left behind. Gizmo path shown in editor." **Assessment:** Output was clean. Ghost trail spawn approach kept as-is. Gizmo gradient (green A to red B) added manually - much more readable than the single-colour gizmos Cursor produced.

[DAY 3 - ENTRY 3]

Type: Polish - UIManager **Entry:** UIManager upgraded with Cursor assistance.

- Death flash: white impact frame (one frame) then cross-fade to red, then slow quadratic fade out
- Camera shake: damped falloff, both cameras, restores to exact origin positions
- Dividing line: triple-flicker with colour shift (white, icy blue, soft yellow) then fade
- Label punch: scale overshoot on SetLevelLabel - makes hint text impossible to miss

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[DAY 3 - ENTRY 4]

Type: Polish - LevelTimer **Entry:** LevelTimer visual upgrades added.

- Inactive timer: icy blue breathing colour, shrinks slightly, vertex wave animation on number characters
- Frost icon: pulses alpha alongside timer breathing
- Critical threshold: red strobe effect (fast sin wave)
- Timer format changed to M:SS.cs for precision

[DAY 3 - ENTRY 5]

Type: Polish - ExitDoor, LevelCompleteUI, HazardTile, CameraFollow **Entry:**

- ExitDoor: impact white frame, settles to green, then continuous pulse. Scale punch on activation.
- LevelCompleteUI: bounce overshoot on slide-in, elastic pop on header text, button fade delay
- HazardTile: white to red flash on contact before kill - visual punch on death
- CameraFollow: predictive look-ahead added - camera anticipates direction of movement

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[DAY 3 - ENTRY 6]

Type: Level Design **Entry:** Levels 5-7 built. Level 5: one-way gates (Platform Effector 2D). Level 6: two switches, order matters. Level 7: all elements combined with timer pressure. All tested and functional.

[DAY 3 - ENTRY 7]

Type: Code - MainMenu **Entry:** MainMenu.cs upgraded.

- Title: fades in, then continuously floats up/down
- Subtitle: fades in after title
- Buttons: slide up from below with fade
- Fade overlay: in on open, out on start
- Buttons require CanvasGroup component for alpha animation

[DAY 3 - ENTRY 8]

Type: Bug Fix - UI Responsive **Entry:** Timer text and frost icons going off screen on resolution changes. Root cause: elements anchored to centre instead of correct screen corners. Fix: ARC timer moved to top-right anchor. ECHO timer moved to bottom-right anchor. Frost icons anchored adjacent to their timers. All other elements confirmed anchored correctly. Canvas Scaler: Scale With Screen Size, 1920x1080, Match 0.5.

Scope Changes Log

Change	Reason	Date
Replaced mirrored simultaneous control	Mirror input was frustrating to play; relay expresses	Day

Change	Reason	Date
with relay mechanic	the theme more directly	1
Level 8 not implemented	7 levels were enough for submission; remaining time went to polish	Day 3
BoostOrb added	Adds strategic depth and cross-character interaction beyond switches	Day 3
MovingHazard added	Adds timing challenge to later levels without redesigning layouts	Day 3

Bug Log

Bug	Status	Fix
Tiles too large in Tilemap	Fixed	Leonardo images resized to 512x512 via birme.net, PPU set to 512
Only seeing blue on play	Fixed	Camera Z set to -10, CameraARC targeted correct world position
Boost duration not expiring	Fixed	Coroutine moved from BoostOrb to PlayerController - survives orb destruction
Y axis double-smoothed in CameraFollow	Fixed	X and Y separated into independent SmoothDamp calls
Timer UI going off screen on resize	Fixed	All elements re-anchored to correct screen corners, Canvas Scaler set to Scale With Screen Size
Characters could move after level complete	Fixed	LevelManager.FreezeAllCharacters() calls SetActive(false) on both after level ends

References

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